

MethylPhys CPG - Operations Manual, chain v3

Development build, not commissioned; not a diagnostic test. Generated 2026-10-03 by manual/build_manual_v3.py from the files named at the start of each part. The procedure is sop/MethylPhys_CPG_SOP_v3.md.

Part 1. Operations - chain v3, neutrophils

Source: manual/OM_v3_neutrophil_chain.md

Operations Manual — chain v3, neutrophils (operator chapter)

Build: development v3 (2026-10-01), not commissioned; this chapter re-checked against the code on 2026-10-03. The procedure is in sop/MethylPhys_CPG_SOP_v3.md; this chapter covers running it and reading the output. The PDF manual in this folder (MethylPhys_CPG_Operations_Manual.pdf) is built from this chapter, the generated chain sequence and the toolkit list by build_manual_v3.py. The class-floor engine (v2) and its manual were retired on 2026-10-03 and are archived privately. Paths are relative to Biological_Physics/MethylPhys/.

Before a run

1. Use EPIC v1 IDAT pairs (Grn and Red), with declared sex and age. Without --sex or --age, Stage 0 quarantines the specimen.
2. Put at least 3 healthy reference specimens of the same specimen type on the same slide (else the same batch), run the same way, so each reading can be tared. This applies to whole blood and to isolated neutrophils.
3. Use Python 3.11 with the versions in chain/requirements.txt (methyldprep 1.7.1, numpy 1.26.4, pandas 1.5.3, scipy 1.17.1; reportlab for the manual): `pip install -r chain/requirements.txt`.
4. Stage 1 needs the Illumina manifest. methyldprep downloads it on first use into `$HOME/.methyldprep_manifest_files/` (HOME must be writable; network access to `array-manifest-files.s3.amazonaws.com`); offline, place the files there by hand (doors/RUNBOOK.md section 1). In a batch, run the first array alone so the download is not raced.
5. The frozen inputs are in the repository under chain/Runtime Matrices/ (Met_A_Floors/, IAM_A_Positions/, Intake/); their hashes are in chain/FROZEN_INPUTS_v3.json.

Run

From chain/MethylPhys_Interface/ (run on the box: Stage 1 needs the manifest):

```
python run_sample.py --grn <Grn> --red <Red> --engine v3 --specimen "whole blood" --array-type EPIC_v1 \
--sex <F|M> --age <years> --id <id> --out <id>.html
```

- Isolated neutrophils: `--specimen "isolated neutrophils"`.
- Pass 2 (the tare): run each specimen again, whole blood and isolated alike, with `--slide-ref-A a1,a2,a3`, where those are the references' untared A values from pass 1, or with `--slide-ref-table refs.csv` (column A, optional column id; a row whose id is the specimen's own --id is left out). Fewer than 3 references: the reading stays untared.
- A beta table already calibrated by this chain's Stage 1 (two-column CSV `cpg_id,beta`; Stage 0 does not run):
`python run_sample.py --betas <table>.csv --specimen "whole blood" --id <id> --out <id>.html`.
- Second draw of the same person (stage 12b): run the first draw with `--save-betas <id>_betas.parquet` `--patient-id <hash>`; run the second with the same `--patient-id` plus `--prior-betas <id>_betas.parquet` `--prior-bundle <id>_bundle.json`.

- Sequencing (Stage Q, IAM-A): `python run_sample.py --pat <file>.pat.gz --id <id> --out <id>.html`, or `python run_sample.py --site-table <sites>.csv --seq-pipeline loyfer_pat_v1 --id <id> --out <id>.html`.

Output: `<id>.html`, `<id>_bundle.json` beside it, and one row appended to `evidence_ledger.jsonl` in the same folder (`--ledger` to change; `--no-bundle` writes neither the bundle nor the row). Exit 0 when a report is written (a refusal still writes one); exit 2 on a Stage 0 QUARANTINE, with no report, no bundle and no ledger row (a command-line error also exits 2).

Batch runners (written for the compute box; they read box paths and roster files and are not general tools):

`chain_tests/run_chain_acceptance.py` (pass 1 every specimen; pass 2 re-runs each whole-blood specimen with the other specimens of its group as references; isolated specimens are not re-run) and

`doors/data/DEV_REPL_V3_01_run/run_proc_repl_v3_01.py` (pass 2 with `--slide-ref-table` built from the other arrays on the same slide, else the batch; exact commands in `COMMANDS.md` beside it).

Checks: `python chain/release_check_v3.py` (or `python kit/release_check.py`) - exit 0 only when every check passes. The IDAT checks E1-E3 need the manifest (run on the box). Manual: `python manual/build_manual_v3.py` rebuilds the PDF.

Reading the report

field	meaning
Refused	the platform check: not an EPIC v1 vector (array type, EPIC v2 probe names, or 700,000 probes or fewer); an EPIC v2 IDAT pair is refused at intake, before Stage 1. No reading
Stage 0 intake	verdict PROCEED / PROCEED_WITH_PENALTY / QUARANTINE (QUARANTINE produces no report), call rate, flags; not run for <code>--betas</code> , <code>--pat</code> , <code>--site-table</code> or <code>--no-intake</code>
Stage 1	poobah detection, call rate and controls: recorded, not gated
Stage A composition	the 8 blood groups (groups at 1 % or more are listed). Whole blood is read when neutrophils are ≥ 20 % and ≥ 867 of the 963 markers are measured
Stage M Met-A	isolated cells: A against the own floor, drawn on the gauge as untared until tared. Whole blood untared: A is a number only, no gauge position. Shift per 1 % loss of the neutrophil pattern
Stage T tare	$A_{rel} = A \div \text{median of the references}$; number of references and their median; reference spread; detection limit (% loss of the pattern) = $2 \times \text{spread} \div \text{shift per 1 \% loss}$. Normal = 0.95–1.05
Noise index N	mean $H(\beta)$ on the 48,528 noise sites. Above 0.149 on an untared reading: state withheld, A printed as a number
Methylated sites mean β	below 0.5: past the entropy ceiling; read β , not A
Stage MC C-score	genomic clustering of the departures (healthy = 1), with the healthy held-out range. Development: no band yet
Stage Q IAM-A	sequencing only: IAM-A, copy error eps, position P, eps0, the two halves, opportunities
Withheld	what the build does not print, and why
Stage 12b difference map	only with <code>--prior-betas</code> / <code>--prior-bundle</code> : per-address difference to an earlier draw of the same person, or the refusal naming what differs (identifier hash, array type, pipeline)
Red flags	STOP / WITHHELD / CAUTION / NOTE, from the bundle; also written to the bundle as <code>red_flags</code>
Safeguards	rendered-claim scan (no disease, cohort or population words in the prose), formula self-test, anchors (Normal band, floor, C baseline, N_{max} read from the frozen files), deconvolver conformance, atlas separability (NOT_RUN until stage 3 is wired)
Troubleshooting, Integrity, Chain file inventory, Run it yourself	what to do for each condition on this specimen; IDAT hashes and chain commit; every frozen input and module with its hash; the exact command
The chain, Toolkit stages	SOP 2b stage table; each toolkit stage PASS / FAIL / REFUSED / NOT_RUN / NOT_BUILT on this specimen

Faults

symptom	cause	action
QUARANTINE_INCOMPLETE_MANIFEST	sex, age or array type not declared (and not readable from the IDAT header)	supply --sex, --age, --array-type
ENVIRONMENT_MISSING_MANIFEST (exit 3; before 2026-10-03 this showed as QUARANTINE_CORRUPT_IDAT) with PermissionError or a connection error on .methylnprep_manifest_files	the manifest is not in its cache and could not be downloaded; the IDAT is not at fault and is not judged	make HOME writable and allow the download, or place the manifest files by hand; re-run
Stage 0 QUARANTINE, other hard failures (call_rate, detection, ctrl_qc, sex, integrity, hm450_coverage, intake_deferred:...)	the array failed intake	none: no reading by rule; the flags name the check
neutrophil fraction <f> < 0.2: fraction reported, A withheld	neutrophils < 20 % of the whole blood	none: the fraction is reported and A is withheld by rule
only <n> of 963 composition markers measured/only <n> of 6000 ... sites measured	fewer than 90 % measured	none: A withheld
untared: <n> same-run reference arrays (>= 3 required)	fewer than 3 references supplied	run the references, then pass them with --slide-ref-A or --slide-ref-table
withheld: noise index <N> > 0.149 and no same-run tare	the array is noisier than the reference arrays; the gauge is not drawn	tare it against same-run references
refusal ... (450K or incomplete vector) ... (450K neutrophil floor pending)	fewer than 700,001 probes after Stage 1 (a 450K table, or an EPIC array that lost probes at detection)	none: v3 reads EPIC v1 only. A 450K IDAT pair never gets here: it quarantines at Stage 0 (0.1 array-type mismatch when --array-type EPIC_v1 is given, else 0.7b coverage)
refusal ... (no frozen neutrophil floor for this platform)	declared array type is not EPIC_v1, or EPIC v2 probe names	none: no floor for that platform
IAM-A refusal position for neutrophils was measured on loyfer_pat_v1, not <pipeline>/too few opportunities	another read-level pipeline, or fewer than 100,000 opportunities	none: P is valid only for its own pipeline

Part 2. The chain, step by step

Source: doors/CHAIN_SEQUENCE.md

The chain, step by step - derived from the code

Generated by chain/build_chain_sequence.py from the AST of run_sample.py, conductor_v3.py and stage_q_iam_a.py. Do not edit by hand: re-run the generator. Every step below is a call the code actually makes, in the order it makes it. Chain v3 is the only engine; the class-floor engine (v2) was retired on 2026-10-03 and is archived privately.

The live path (chain v3) - 24 steps, in order

#	step	implemented in	what it does
1	step_0_1_idat_arrival	stage_0_intake.py	SOP section 0.1_idat_arrival
2	step_0_2_manifest_creation	stage_0_intake.py	SOP section 0.2_manifest_creation
3	step_0_3_integrity_hash	stage_0_intake.py	SOP section 0.3_integrity_hash
4	step_0_4_control_probe_validation	stage_0_intake.py	SOP section 0.4_control_probe_validation
5	step_0_5_detection_pvalue_qc	stage_0_intake.py	SOP section 0.5_detection_pvalue_qc
6	step_0_6_bead_count_qc	stage_0_intake.py	SOP section 0.6_bead_count_qc
7	step_0_7_call_rate	stage_0_intake.py	SOP section 0.7_call_rate
8	step_0_7b_platform_coverage	stage_0_intake.py	SOP section 0.7b_platform_coverage
9	step_0_8_sex_check	stage_0_intake.py	SOP section 0.8_sex_check
10	step_0_9_decision_gate	stage_0_intake.py	SOP section 0.9_decision_gate
11	Stage 1 - IDAT calibration	stage_1_idat_calibration.py	Stage 1 - IDAT calibration to beta (SOP Stage 1, steps 1.1-1.2 + 1.5) _called by run_sample.py when given --grn/--red; skipped when given --betas_
12	platform_refusal	conductor_v3.py	None when the specimen is EPIC v1; otherwise the refusal text
13	stage_m_isolated	conductor_v3.py	Isolated / sorted neutrophils against their own floor. The own-floor state is kept as state_own_floor; the reading's state is
14	stage_a_composition	conductor_v3.py	EPIC blood composition (blood_composition_EPIC_v1): 8 groups from Salas purified EPIC cells; markers exclude the neutrophil sites;
15	stage_m_blood	conductor_v3.py	Whole blood: Met-A = mean H(beta) / mean H(e) at the neutrophil sites, e = sum_g f_g mu_g (EPIC purified group profiles)
16	noise_index	conductor_v3.py	N = mean H(beta) over the noise sites measured on this array; None when fewer than MIN_NOISE_FRACTION are measured _stage 9: the array's noise index N over the noise sites_
17	stage_t_tare	conductor_v3.py	Same-run tare against >= MIN_REFS healthy references of the same specimen type run the same way (same slide, else same batch)

#	step	implemented in	what it does
18	noise_gate	conductor_v3.py	Runtime Matrices/Met_A_Floors/noise_gate_EPIC_v1.json: N_max = top of the reference arrays' noise range (DEV-NOISE-01) _stage 9: N_max read here; run_neutrophil withholds the gauge state when N > N_max and the reading is untared_
19	stage_mc_cscore	conductor_v3.py	Met-A C-score (stage 6): clustering of the residual z map in genomic order - variance of the means of blocks of clustering_block
20	stage_q_iam_a.pat_site_table	stage_q_iam_a.py	Per-site table (pos, opp_A, err_A, opp_B, err_B) from one .pat(.gz) file, pipeline loyfer_pat_v1. pos = 'chrom:CpG index' _runs when run_sample.py is given --pat or --site-table_
21	stage_q_iam_a.read	stage_q_iam_a.py	site_table: per-site table (see module doc). cell: e.g. 'neutrophils'. pipeline: the read-level pipeline that produced the table _runs when run_sample.py is given --pat or --site-table_
22	serial_mode.check_same_person	serial_mode.py	Refuse unless patient hash, array type and pipeline match. Returns (ok, reason) _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
23	serial_mode.delta_sky	serial_mode.py	beta_now - beta_prior per address on the addresses both draws measured. No expectation, no sigma, no composition _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
24	Report	MethylPhys_Interface/repor_t_v3.py	one self-contained HTML page plus the JSON bundle

Stage modules in chain/ that the live path does not call

module	what it implements	status
stage_2c_trace_detection.py	Stage 2c - is there any evidence of a trace class in this specimen?	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
stage_4_6_patient_cmb.py	Stage 4.6 — the patient's sky (PROC-CMB-01, 2026-09-21). Replaces the retired patient_brightness_comparison.py	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
stage_a_composition_v2.py	Stage A - composition, atlas v2 solver (deconv_v2). Not called by chain v3 (conductor_v3.stage_a_composition uses blood_composition_EPIC_v1.json). Frozen settings (commissioned as PROC-DECONV-V2-01; specimen cell sets 2026-09-30)	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)

The toolkit (chain/TOOLKIT.md) - built; each stage wired only after its check passed

stage (SOP 2b)	name	module	status
3	Atlas deconvolution	chain/deconv_v2.py	toolkit: not yet wired into chain v3
3b	Trace-cell detection	chain/stage_2c_trace_detection.py	toolkit: not yet wired into chain v3

stage (SOP 2b)	name	module	status
3c	Foreign-cell detection	chain/toolkit_foreign_detection.py	toolkit: not yet wired into chain v3
4	NILC component separation	chain/nilc_celltype_deconvolver.py	toolkit: not yet wired into chain v3
4	NILC helper: lineage splitter	chain/Lineage_Splitter/lineage_splitter.py	toolkit: not yet wired into chain v3
10	Directional decomposition	chain/Runtime Matrices/Directional Panel/bidirectional_decomposition.py	toolkit: not yet wired into chain v3
11	Sky map	chain/stage_4_6_patient_cmb.py	toolkit: not yet wired into chain v3
11b	Surface brightness	chain/toolkit_surface_brightness.py	toolkit: not yet wired into chain v3
12	Sky statistics	chain/sky_statistics.py	toolkit: not yet wired into chain v3
12b	Difference map	chain/serial_mode.py	wired into chain v3 (optional input; see chain/TOOLKIT.md)

Part 3. The toolkit

Source: chain/TOOLKIT.md

chain/TOOLKIT.md - the toolkit: built, kept, entering chain v3 one stage at a time (commissioning 2026-10-03: 12b wired; the rest not)

Every module below is in the repository and importable. **None of it is part of a reading**; stage 12b (difference map) is wired as an optional second-draw comparison and changes no reading. Each enters chain v3 at commissioning, one stage at a time, after its own pre-registered check has passed on v3 and the outcome is recorded in doors/ (order and checks: . . /sop/MethylPhys_CPG_SOP_v3.md section 2b). Every module carries the line *Toolkit: not yet wired into chain v3; enters the chain at commissioning with its own pre-registered check*. Stage numbers and names are those of SOP v3 section 2b. Paths are relative to Biological_Physics/MethylPhys/. The class-era engine these modules were first used with (chain v2) was retired on 2026-10-03 and is archived privately; nothing here imports it.

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status
3	Atlas deconvolution	chain/deconv_v2.py (DeconvV2); wrapper chain/stage_a_composition_v2.py; development copies atlas/v2/postbuild/scripts/deconv_v2.py, methylphys_v2.py	one joint non-negative solve over the atlas v2 cell types, $f \geq 0$, $\sum f = 1$, inverse-variance weights iterated, markers chosen per cell against its nearest cells; presence by bootstrap of the marker loci	the atlas v2 reference (parquet, built by atlas/v2/scripts; not stored in the repository), a beta vector on the Stage 1 scale; frozen solver settings in stage_a_composition_v2.py	check failed 2026-10-03 (DEV-ATLAS-EPIC-02: held-out truth bars failed; rule and repeatability and agreement passed) - not wired
3b	Trace-cell detection	chain/stage_2c_trace_detection.py; panel chain/Runtime Matrices/trace_detection_panel_v1.json	inverse-variance score test below the non-negativity boundary: fit without the candidate, project its profile off the fitted columns, regress the residual on it unconstrained	a beta vector; the panel (addresses, per-class means and SDs, thresholds)	not run 2026-10-03 (DEV-TOOLKIT-ADDED-01: line must be re-set without a population first) - not wired; author decision
3c	Foreign-cell detection	chain/toolkit_foreign_detection.py (stage_2d_foreign_detection); panel chain/Runtime Matrices/Celltype_Marker/detection_panel_v3.json	one joint NNLS of the panel markers on [blood reference columns, foreign-cell templates]; each template's fraction read against its measured noise floor; three or more above floor = "material not resolved"	a beta vector, the panel, optional composition-guard record	not run 2026-10-03 (DEV-TOOLKIT-ADDED-01: line is a 732-array quantile; class-era beta scale) - not wired; author decision
4	NILC component separation	chain/nilc_celltype_deconvolver.py (NILCCelltypeDeconvolver)	internal linear combination per cell type: variance-weighted generalised least squares over the markers, ridge-regularised, bootstrap CI per fraction	atlas CSV (atlas/IA MatlasREBUILD.csv.xz, decompressed), marker file chain/Runtime Matrices/Celltype_Marker/iamatlas_celltype_markers_v0_2.json, a beta vector	check failed 2026-10-03 (DEV-NILC-01: both the module and NILC-e failed the held-out truth bars) - not wired

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status
4	NILC helper: lineage splitter	chain/Lineage_Splitter/lineage_splitter.py (LineageSplitter)	splits one compartment's mass among its member classes by a small NNLS on the contrast CpGs ($\text{abs}(\text{mean}_a - \text{mean}_b) \geq \text{min_delta}$, weighted by the contrast), with a condition-number check; reports the compartment only when the split is ill-conditioned	atlas CSV, a beta vector, the compartment fraction and its members	helper of stage 4; not wired (stage 4 check failed)
10	Directional decomposition	chain/Runtime Matrices/Directional_Panel/bidirectional_decomposition.py; panel_directional_panels_v1_0.json beside it	per-CpG signed z against the panel's frozen direction, summed into a directional composite beside the pooled-entropy reading, so a departure's sign (toward disorder or toward over-order) is kept	a beta vector, the directional panel	not run 2026-10-03 (DEV-DIRECTION-01: cohort-trained disease-direction panel; output cannot express disorder / over-order) - not wired; author decision
11	Sky map	chain/stage_4_6_patient_cmb.py (patient_sky_sigma, render_plate); mapping chain/Runtime Matrices/Patient_CMB/iamatlas_cpg_to_healpix_nside128.npz built by chain/Runtime Matrices/Patient_CMB/build_healpix_mapping.py; earlier mapping atlas/healpix_mapping/	residual $r = \text{beta} - \text{sum f mu}$ per CpG, scaled by the propagated atlas posterior and the array's SNP-probe noise, projected onto HEALPix NSIDE 128 (RING) in genomic order; panels masked below the presence floor; Mollweide plate	a beta vector, fractions, atlas means and SDs (atlas CSV), the mapping, chain/Runtime Matrices/Patient_CMB/presence_floors_v1.json	check failed 2026-10-03 (DEV-SKY-01: healthy replicate skies 1.1-6x their shuffled null at $l < 192$) - not wired
11b	Surface brightness	chain/toolkit_surface_brightness.py (attach_brightness_ci)	95 % interval on a reading by Monte Carlo through $A = H(\text{beta_mean}) / H_{\text{min}}$, using the atlas per-CpG brightness posterior SD at the scored loci	atlas/iamatlas_class_archives/*_REBUILD.tar.xz (per-CpG brightness CSVs), the scored loci and floor	not run 2026-10-03 (DEV-TOOLKIT-ADDED-01: class-era inputs) - not wired; author decision
12	Sky statistics	chain/sky_statistics.py (masked_spectrum, bandpowers, shuffled_null); null suite chain/CPG_Null_Runner/cpg_null_runner.py; per-run tool registry chain/cmb_tools.py	masked pseudo angular power spectrum (healpy anafast) in six bands; spatially shuffled null (pixel values permuted within the mask); label-permutation, split, injection and look-elsewhere (Bonferroni, null N8) nulls; a registry that marks each borrowed tool PASS / FAIL / NOT_RUN / NOT_BUILT on a bundle	a HEALPix residual map (stage 11); for the null suite, a procedure's per-sample readings; for the registry, a run bundle	not run 2026-10-03 (DEV-SKYSTAT-01: look-elsewhere by simulation not built; stage 11 upstream open) - not wired

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status
12b	Difference map	chain/serial_mode.py (check_same_person, delta_sky, delta_cells, trajectory)	per-address beta difference of two draws of one person on the addresses both measured, refused unless both draws share the identifier hash, array type and pipeline; per-cell A differences across draws (design doors /PROC_SERIAL_01_PREREG.md)	two runs (bundles and beta vectors) of one person	wired 2026-10-03 behind --prior-betas/--prior-bundle (DEV-TOOLKIT-ADD ED-01 passed; release check E5): per-address difference + same-person check. Not built: the difference drawn as a sky; delta_cells/trajectory (class-era fields) not called

Notes

- CPG_Null_Runner/cpg_null_runner.py null N7 loads the synthetic patient generator, which was retired with chain v2; N7 reports DEFERRED until a v3 generator exists. The other nulls run without it.
- cmb_tools.py checks fields of the retired v2 bundle (patient_sky, foreign_detection, second_opinion); on a v3 bundle every entry reads NOT_RUN until the stage it checks is wired in.
- The class-era wrappers that called these modules inside the retired conductor (the stage 4.6 sky wrapper, the class-level NILC second opinion) were not carried over: they read retired files (class identity loci, beta scale maps). Each stage gets a v3 wrapper when it is commissioned.

Sources

manual/OM_v3_neutrophil_chain.md sha256 667811e36543

doors/CHAIN_SEQUENCE.md sha256 235bb0def69c

chain/TOOLKIT.md sha256 dce9d3d505b1